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1  /* =====
2  3. TRANSFER_MONEY
3  Transfers between two active accounts in the banking system
4  ===== */
5
6  CREATE OR ALTER PROCEDURE Transfer_Money
7  (
8      @FromAccountID INT,
9      @ToAccountID INT,
10     @TransferAmount DECIMAL(18,2),
11     @Category VARCHAR(30),
12     @Channel VARCHAR(20)
13 )
14 AS
15 BEGIN
16     SET NOCOUNT ON;
17
18     BEGIN TRY
19         BEGIN TRANSACTION;
20
21         -- 1. Validate transfer amount
22         IF @TransferAmount <= 0
23         BEGIN
24             THROW 50001, 'Transfer amount must be greater than zero.', 1;
25         END;
26
27         -- 2. Sender and receiver cannot be the same account
28         IF @FromAccountID = @ToAccountID
29         BEGIN
30             THROW 50002, 'Sender and receiver accounts cannot be the same.', 1;
31         END;
32
33         -- 3. Validate transfer category
34         IF @Category NOT IN ('UPI', 'IMPS', 'NEFT', 'RTGS')
35         BEGIN
36             THROW 50003, 'Invalid transfer category.', 1;
37         END;
38
39         -- 4. Validate sender account
40         IF NOT EXISTS
41         (
42             SELECT 1
43             FROM Accounts
44             WHERE AccountID = @FromAccountID
45                   AND AccountStatus = 'Active'
46         )
47         BEGIN
48             THROW 50004, 'Sender account does not exist or is not active.', 1;
49         END;
50
51         -- 5. Validate receiver account
52         IF NOT EXISTS
53         (
54             SELECT 1
55             FROM Accounts
56             WHERE AccountID = @ToAccountID
57                   AND AccountStatus = 'Active'
58         )
59         BEGIN
60             THROW 50005, 'Receiver account does not exist or is not active.', 1;
61         END;
62
63         -- 6. Validate Category + Payment Channel
64         IF
65         (
66             (@Category = 'UPI' AND @Channel <> 'Mobile Banking')
67             OR (@Category = 'IMPS' AND @Channel NOT IN ('Branch', 'Internet Banking', 'Mobile Banking'))
68             OR (@Category = 'NEFT' AND @Channel NOT IN ('Branch', 'Internet Banking', 'Mobile Banking'))
69             OR (@Category = 'RTGS' AND @Channel NOT IN ('Branch', 'Internet Banking', 'Mobile Banking'))
70         )
71         BEGIN
72             THROW 50006, 'Invalid Category and Payment Channel combination.', 1;
73         END;
74     END TRY
75     BEGIN CATCH
76         IF @@TRANCOUNT > 0
77             ROLLBACK TRANSACTION;
78     END CATCH;
79 END;

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68 -- 7. Validate Mobile Banking
69 IF @Channel = 'Mobile Banking'
70 BEGIN
71     IF NOT EXISTS
72     (
73         SELECT 1
74         FROM DigitalBanking
75         WHERE AccountID = @FromAccountID
76               AND MobileBankingEnabled = 1
77     )
78     BEGIN
79         THROW 50007, 'Mobile Banking is not enabled for the sender account.', 1;
80     END;
81 END;
82
83 -- 8. Validate Internet Banking
84 IF @Channel = 'Internet Banking'
85 BEGIN
86     IF NOT EXISTS
87     (
88         SELECT 1
89         FROM DigitalBanking
90         WHERE AccountID = @FromAccountID
91               AND InternetBankingEnabled = 1
92     )
93     BEGIN
94         THROW 50008, 'Internet Banking is not enabled for the sender account.', 1;
95     END;
96 END;
97
98 -- 9. Validate UPI
99 IF @Category = 'UPI'
100 BEGIN
101     IF NOT EXISTS
102     (
103         SELECT 1
104         FROM DigitalBanking
105         WHERE AccountID = @FromAccountID
106               AND UPIEnabled = 1
107     )
108     BEGIN
109         THROW 50009, 'UPI is not enabled for the sender account.', 1;
110     END;
111 END;
112
113 -- 10. Get sender balance
114 DECLARE
115     @FromOpeningBalance DECIMAL(18,2),
116     @FromAvailableBalance DECIMAL(18,2),
117     @OverdraftLimit DECIMAL(18,2);
118
119 SELECT
120     @FromOpeningBalance = CurrentBalance,
121     @FromAvailableBalance = AvailableBalance,
122     @OverdraftLimit = OverdraftLimit
123 FROM Accounts WITH (UPDLOCK, HOLDLOCK)
124 WHERE AccountID = @FromAccountID;
125
126 -- 11. Validate sender funds
127 IF (@FromAvailableBalance + @OverdraftLimit) < @TransferAmount
128 BEGIN
129     THROW 50010, 'Insufficient funds for this transfer.', 1;
130 END;
131
132 -- 12. Get receiver opening balance
133 DECLARE @ToOpeningBalance DECIMAL(18,2);
134
135 SELECT @ToOpeningBalance = CurrentBalance
136 FROM Accounts WITH (UPDLOCK, HOLDLOCK)
137 WHERE AccountID = @ToAccountID;
138
139 -- 13. Get sender and receiver account numbers
140 DECLARE
141     @FromAccountNumber VARCHAR(20),
142     @ToAccountNumber VARCHAR(20);
143
144 SELECT @FromAccountNumber = AccountNumber
145 FROM Accounts WHERE AccountID = @FromAccountID;
146 SELECT @ToAccountNumber = AccountNumber
147 FROM Accounts WHERE AccountID = @ToAccountID;

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-- 14. Generate common transfer reference
DECLARE @TransferReferenceNumber VARCHAR(30);

SET @TransferReferenceNumber =
    'TRF-'
    + CONVERT(VARCHAR(8), CAST(SYSDATETIME() AS DATE), 112)
    + 'Aa'
    + RIGHT('000000' + CAST(NEXT VALUE FOR TransactionReferenceSeq AS INT), 6);

-- 15. Debit sender account
UPDATE Accounts
SET
    CurrentBalance = CurrentBalance - @TransferAmount,
    AvailableBalance = AvailableBalance - @TransferAmount,
    LastTransactionDate = SYSDATETIME(),
    ModifiedDate = SYSDATETIME()
WHERE AccountID = @FromAccountID;

-- 16. Record sender debit transaction
INSERT INTO Transactions
(
    TransactionReferenceNumber,
    TransferReferenceNumber,
    AccountID,
    TransactionType,
    TransactionCategory,
    PaymentChannel,
    Amount,
    Charges,
    TaxAmount,
    OpeningBalance,
    ClosingBalance,
    CounterpartyAccountNumber,
    CounterpartyName,
    CounterpartyBank,
    TransactionStatus,
    Remarks,
    InitiatedBy,
    ApprovedBy,
    TransactionDate,
    CreatedDate
)
VALUES
(
    @TransferReferenceNumber + '-D',
    @TransferReferenceNumber,
    @FromAccountID,
    'Debit',
    @Category,
    @Channel,
    @TransferAmount,
    0.00,
    0.00,
    @FromOpeningBalance,
    @FromOpeningBalance - @TransferAmount,
    @ToAccountNumber,
    NULL,
    NULL,
    'Success',
    @Category + ' transfer sent via ' + @Channel,
    NULL,
    NULL,
    SYSDATETIME(),
    SYSDATETIME()
)

```

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--Example : Transfer money through UPI using Mobile Banking
EXEC Transfer_Money
    @FromAccountID = 11, @ToAccountID = 35, @TransferAmount = 2000, @Category = 'UPI', @Channel = 'Mobile Banking';
--Example : Transfer money through IMPS using Internet Banking
EXEC Transfer_Money
    @FromAccountID = 44, @ToAccountID = 33, @TransferAmount = 10000, @Category = 'IMPS', @Channel = 'Internet Banking';
--Example : Transfer money through NEFT using Branch
EXEC Transfer_Money
    @FromAccountID = 46, @ToAccountID = 61, @TransferAmount = 25000, @Category = 'NEFT', @Channel = 'Branch';
--Example : View transfer transactions for selected accounts
select * from Transactions
where AccountID IN(11,35,44,33,46,61)
--Example : View updated balances for selected accounts
select * from Accounts
where AccountID IN(11,35,44,33,46,61)
```

Before
Money Transfer

TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remarks
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TransferReferenceNumber	FromAccountID	ToAccountID	TransferAmount	SenderClosingBalance	ReceiverClosingBalance	TransferCategory	PaymentChannel	Message
TRF-20260816-000053	11	35	2000.00	71257.38	589581.11	UPI	Mobile Banking	Transfer successful.
TRF-20260816-000054	44	33	10000.00	476534.48	824810.00	IMPS	Internet Banking	Transfer successful.
TRF-20260816-000055	46	61	25000.00	611089.33	550643.41	NEFT	Branch	Transfer successful.

After
Money Transfer

TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remarks
52	TRF-20260816-000053-D	TRF-20260816-000053	11	Debit	UPI	Mobile Banking	2000.00	0.00	0.00	73257.38	71257.38	BSGB019000035	NULL	NULL	Success	UPI tra
53	TRF-20260816-000053-C	TRF-20260816-000053	35	Credit	UPI	Mobile Banking	2000.00	0.00	0.00	587581.11	589581.11	BSGB017000011	NULL	NULL	Success	UPI tra
54	TRF-20260816-000054-D	TRF-20260816-000054	44	Debit	IMPS	Internet Banking	10000.00	0.00	0.00	486534.48	476534.48	BSGB009000033	NULL	NULL	Success	IMPS tr
55	TRF-20260816-000054-C	TRF-20260816-000054	33	Credit	IMPS	Internet Banking	10000.00	0.00	0.00	814810.00	824810.00	BSGB011000044	NULL	NULL	Success	IMPS tr
56	TRF-20260816-000055-D	TRF-20260816-000055	46	Debit	NEFT	Branch	25000.00	0.00	0.00	636089.33	611089.33	BSGB010000061	NULL	NULL	Success	NEFT tr
57	TRF-20260816-000055-C	TRF-20260816-000055	61	Credit	NEFT	Branch	25000.00	0.00	0.00	525643.41	550643.41	BSGB015000046	NULL	NULL	Success	NEFT tr

AccountID	AccountNumber	CustomerID	BranchID	AccountTypeID	OpeningDate	CurrentBalance	AvailableBalance	LienAmount	OverdraftLimit	CurrencyCode	AccountStatus	FreezeReason	ClosedDate	ClosedByEmployeeID	LastTransactionDate	CreatedDate	ModifiedDate
11	BSGB017000011	15	17	1	2021-09-13	71257.38	71257.38	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-16 12:33:03.7749186	2026-08-09 17:07:30.1766834	2026-08-16
33	BSGB009000033	30	9	1	2020-04-14	824810.00	824810.00	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-16 12:33:03.8088363	2026-08-09 17:07:30.1766834	2026-08-16
35	BSGB019000035	32	19	1	2021-06-16	589581.11	589581.11	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-16 12:33:03.8058365	2026-08-09 17:07:30.1766834	2026-08-16
44	BSGB011000044	39	11	2	2021-11-13	476534.48	476534.48	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-16 12:33:03.8078310	2026-08-09 17:07:30.1766834	2026-08-16
46	BSGB015000046	41	15	6	2019-01-27	611089.33	611089.33	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-16 12:33:03.8098241	2026-08-09 17:07:30.1766834	2026-08-16
61	BSGB010000061	53	10	1	2020-02-09	550643.41	550643.41	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-16 12:33:03.8108230	2026-08-09 17:07:30.1766834	2026-08-16